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Some Preliminary Measurements of Anisotropy of Magnetic Susceptibility of Rocks

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Summary

Measurements have been made on four sets of rocks, two of which showed anisotropy of magnetic susceptibility and two were isotropic. The maximum and intermediate principal susceptibilities lie in the cleavage plane for some Welsh slates and in the horizontal plane for some igneous rocks of the Skaergaard layered intrusion. In both cases the directions of natural remanent magnetization have been affected. The two sets of isotropic rocks are from the northern Pyrenees and Deccan of India respectively. Both may have been affected by stress environments, the nature of which did not cause measurable anisotropy. It is unlikely that the directions of natural remanent magnetization have been affected for these rocks.

1. Introduction

Apart from the classical work of Ising (1943) on the susceptibility anisotropy of the varve clays of Sweden, few detailed studies have been made of the anisotropy of magnetic susceptibility of rocks.

Graham (1954) found the maximum susceptibility lies in the bedding plane for some sediments but perpendicular to the bedding plane for some folded ferruginous sandstones. He also found some diabase dykes which showed 0·1 to 10 per cent anisotropy. Howell & others (1958) obtained similar results for sediments and found the plane of maximum susceptibility to lie in the foliation plane for some metamorphic rocks. Hargraves (1959) made a study of the susceptibility anisotropy and remanent magnetization of the Allard Lake haemo-ilmenite deposits of Quebec and found the plane of maximum susceptibility coincides with the preferred crystallographic grain orientation of titan-hematite and the remanent magnetization lies in or near this plane.

At the Department of Geodesy and Geophysics, Cambridge, measurements of susceptibility anisotropy have been made on several different rock types using an a.c. bridge. With this apparatus differences of susceptibility of 2 per cent may be reliably measured. The experimental method and computational procedure have been described separately (Girdler 1961). For small fields, the anisotropy of susceptibility may be represented by an ellipsoid, the lengths of the principal axes being proportional to the maximum, intermediate and minimum principal susceptibilities. To help visualize the degree of anisotropy the results are presented by giving the ratios of the *maximum* : *intermediate* : *minimum* susceptibilities. Throughout, special reference is made to the effect of susceptibility anisotropy on the directions of natural remanent magnetization.

2. Welsh slates

The susceptibility anisotropy of two specimens of Welsh slates of Ordovician age was measured and the ratios of maximum/minimum and intermediate/minimum susceptibilities are given in Table 1. The directions of the three principal susceptibilities are shown in relation to the rock cleavage on Schmidt Equal area projections in Figures 1(a) and 1(b). The maximum and intermediate susceptibilities are found to lie close to the cleavage plane and the minimum susceptibility is at right

Table 1

Susceptibility anisotropy of Welsh slates

Specimen No.	Ratio Max. : Int. : Min.	χ_{min} (e.m.u./g)
D–11–1	2·28 : 1·63 : 1	$2{\cdot}58\times10^{-4}$
D–14–1	1·80 : 1·44 : 1	$2{\cdot}69\times10^{-4}$

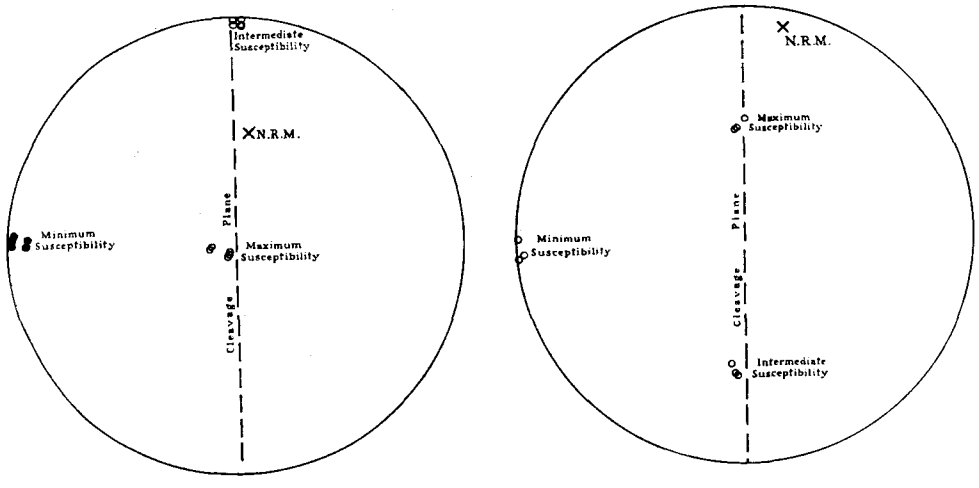


FIG. 1.—Directions of principal susceptibilities and natural remanent magnetization for some Welsh slates.
The plane of the projection (Schmidt Equal Area) is horizontal; positive directions (downwards) are shown by dots and negative directions (upwards) are shown by open circles.

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angles to the cleavage. The natural remanent magnetization was measured using an astatic magnetometer and was found to lie within 10° of the cleavage for both specimens. This work has been continued by Fuller who has obtained similar results for 50 specimens.

X-ray and Curie temperature studies (Fuller 1960) indicate that the main ferromagnetic constituent is pyrrhotite ($Fe_{1-x}S$) for slates with ratio of maximum/minimum susceptibility greater than 1·4. Microscopic examination showed the pyrrhotite to be in the form of platey grains lying parallel to the cleavage. As pyrrhotite has an easy plane of magnetization the observed anisotropy of the slates is most likely due to the alignment of pyrrhotite grains with their high susceptibility basal planes parallel to the cleavage.

3. Skaergaard rocks

The anisotropy of susceptibility has been measured for seven orientated samples from the Skaergaard intrusion of east Greenland, the structure and petrology of which have been described by Wager & Deer (1939). The rocks are ferro-gabbros, which contain pyroxenes and olivines abnormally rich in combined iron. The

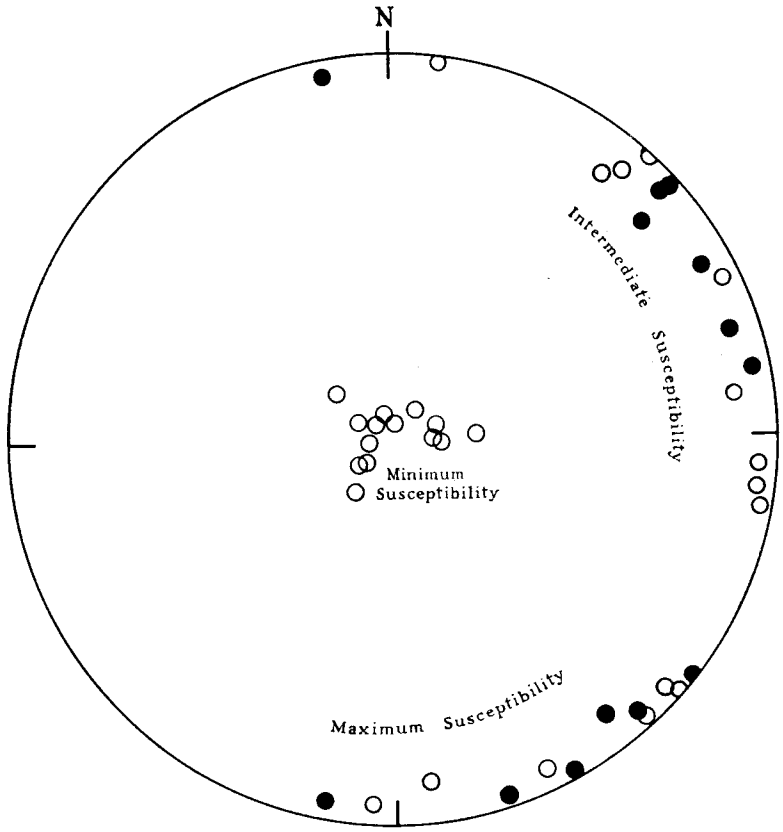


FIG. 2.—Directions of principal susceptibilities for Skaergaard Intrusive rocks. The plane of the projection (Schmidt Equal Area) is horizontal; positive directions (downwards) are shown by filled circles and negative directions (upwards) are shown by open circles.

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Skaergaard intrusion is famous for its gravitational layering and the samples studied showed typical layered structure.

Fourteen cylinders were machined from samples provided by Professor Wager. For all specimens the minimum susceptibility is found to differ appreciably from the maximum and intermediate susceptibilities. The difference between maximum and intermediate susceptibilities is small. The mean differences are: maximum ~ minimum 8·6 per cent, intermediate ~ minimum 6·2 per cent, and maximum ~ intermediate 2·6 per cent. The smallest value of χ is $1\,450 \times 10^{-6}$ e.m.u./g.

The directions of the principal susceptibilities are given in Table 2, where the mean directions have been computed using the method described elsewhere by the author (Girdler 1961). Figure 2 shows the directions of the maximum,

Table 2

Directions of principal susceptibilities of Skaergaard Rock specimens

	Maximum		Intermediate		Minimum	
	θ	ϕ	θ	ϕ	θ	ϕ
3804A	138·6	+5·8	48·0	+6·1	91·4	−81·6
3804B	128·9	+1·0	39·1	−13·9	215·0	−76·1
3804C	131·9	−1·6	41·7	−9·4	231·4	−80·5
Mean	133·0	+2·1	43·2	−5·4	202·1	−84·2
3805A	175·8	−11·0	83·6	−11·2	309·5	−74·3
3805B	151·9	+1·9	61·6	+9·4	73·2	−80·4
Mean	169·8	−7·0	79·7	−0·8	342·9	−82·3
3806	155·8	−6·2	65·2	−5·4	294·2	−81·9
3807A	142·7	+10·5	49·9	+14·4	87·7	−72·1
3807B	162·9	+3·3	72·4	+8·6	93·9	−80·8
3807C	80·2	+5·8	349·7	+5·3	37·5	−82·2
Mean	155·8	+2·1	65·4	+10·3	77·2	−79·4
3808A	133·2	−4·4	43·1	−0·6	305·7	−85·6
3808B	138·1	−2·3	48·2	+2·4	3·5	−86·7
Mean	136·5	−3·3	46·6	−0·8	329·8	−86·6
3809	98·1	−5·3	7·9	−1·8	259·3	−84·4
3811A	191·4	−4·4	101·1	−3·4	334·1	−84·5
3811B	183·9	+5·8	94·5	−5·6	228·2	−82·0
Mean	186·6	+1·5	96·9	−4·2	257·8	−85·5

θ = Declination, measured clockwise from north.
ϕ = Inclination, with reference to the horizontal plane (positive inclination—downwards, negative inclination—upwards).

intermediate and minimum susceptibilities plotted on a Schmidt Equal Area projection. It is seen that the minimum susceptibilities are close to the vertical and the maximum and intermediate susceptibilities are all close to the horizontal plane but are widely scattered within it. Although the maximum and intermediate susceptibilities show considerable scatter, they lie within 65° of arc, the maximum in the south-east quadrant and the intermediate in the north-east quadrant.

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The scatter of the directions of principal susceptibilities is related to the differences between their magnitudes. Since the minimum susceptibility is considerably different from both the maximum and intermediate susceptibilities, the directions of minimum susceptibility are closely grouped; but since the magnitudes of the maximum and intermediate susceptibilities are nearly equal, their directions are widely scattered in the horizontal plane.

The susceptibility anisotropy is consistent with the petrology and fabric of the rocks. Vincent & Phillips (1954) have studied the opaque minerals and Chevallier & others (1954) show that cubic titano-magnetite is the main ferromagnetic constituent. As titano-magnetite is magnetically isotropic, the observed anisotropy should be due to the shape and distribution of these grains. Plate I shows photographs of four polished sections. (a), (b) and (c) are horizontal sections and (d) is a vertical section. As titano-magnetite has high reflectivity, the grains appear white and their shape can be easily studied. The horizontal sections reveal that the titano-magnetites are approximately equidimensional in the horizontal plane but the vertical section shows that the grains are flattened and may be distributed in layers. Measurements have been made of the grain shape (Uyeda & others 1961) and it was found that the observed anisotropy is in excellent agreement with that calculated from the shape of the grains. It is concluded that the susceptibility anisotropy of the Skaergaard rocks is due to the preferred alignment of slightly oblate titano-magnetite grains in the horizontal plane.

The natural remanent magnetization was measured for the fourteen specimens and all were found to be reversely magnetized. The mean directions for each sample are shown by open circles in Figure 3.

Uyeda (see Uyeda & others 1961), carried out a series of heating experiments on a set of magnetite disks and on one of the Skaergaard specimens. The specimens were heated in a nitrogen furnace to above their Curie temperatures and cooled in the present geomagnetic field. For the magnetite disks, the thermal remanent magnetization deviates from the direction of the external field towards the plane of maximum susceptibility, the deviation increasing as the shape factor increases. A similar effect was observed for the Skaergaard specimen, the deviation from the ambient field being approximately 5°. Thus it was found that the directions of n.r.m. do not represent the direction of the Earth's magnetic field at the time of cooling of the rock.

The deviation can also be studied by a graphical consideration of the directions of natural remanent magnetization and principal susceptibilities of all the samples. It has been shown experimentally that the direction of thermal remanent magnetization migrates away from the direction of the ambient field towards the plane of maximum susceptibility. Therefore, it may be assumed that a great circle passing through the directions of remanent magnetization and minimum susceptibility should pass through the ambient field direction. As the degree of preferred orientation is different for each sample the effect will differ and a set of great circles will be obtained which should ideally intersect at a common point which is the direction of the ambient field at the time of cooling of the rock. This has been done in Figure 3 where the directions of n.r.m. and minimum susceptibility for each sample have been joined by great circles. The mean directions and 95 per cent circles of confidence (Fisher 1953) have been computed for the directions of n.r.m., minimum susceptibility and intersections of the great circles and these are illustrated in Figure 4. It is seen that the mean direction of natural remanent magnetization differs from the direction of the ambient field by

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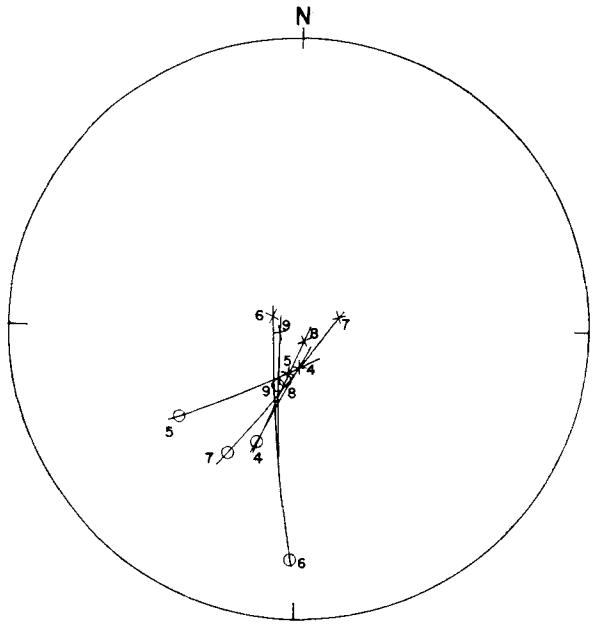


FIG. 3.—Great circles joining mean directions of natural remanent magnetization and mean directions of minimum susceptibility for each sample.

The plane of the projection (Schmidt Equal Area) is horizontal; directions of n.r.m. (negative) are shown by open circles and directions of minimum susceptibility (upwards) are shown by crosses.

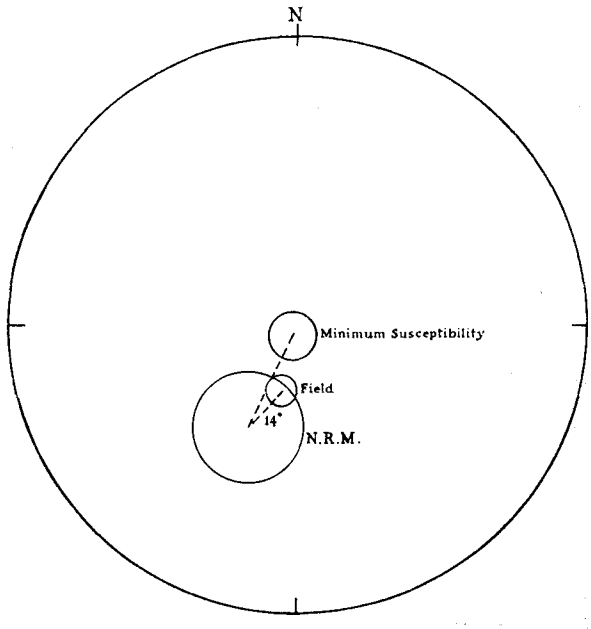


FIG. 4.—Relationship between natural remanent magnetization, minimum susceptibility and ambient field for Skaergaard rocks.

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approximately 14°. It is also noted that although the difference between minimum and maximum susceptibilities is sufficient to cause a deviation of the direction of n.r.m. from that of the ambient field, it seems from Figure 4 that the difference between maximum and intermediate susceptibilities is too small to cause any appreciable effect.

4. Lower Jurassic volcanic rocks from the Northern Pyrenees

The susceptibility anisotropy has been measured for 25 samples of some igneous rocks from the northern foothills of the Pyrenees. These were chosen as part of the preliminary study to see if igneous rocks which have been subjected to orogenesis show preferred directions of susceptibility anisotropy. If any preferred directions exist, it seemed possible that these would be related to the Alpine fold directions of the Pyrenees.

Samples were collected from two areas, the first is located west of Foix (latitude 42° 57′ N, longitude 1° 35′ E) and the second is south-east of Bayonne (latitude 43° 30′ N, longitude 1° 28′ W). The relevant geology of the region is summarized in Girdler (1958) where the directions of natural remanent magnetization for rocks from the first locality are given. The age of the rocks is Lower Hettangian, i.e. equivalent to the Upper Lias of the English Jurassic System.

The rocks collected from four sites west of Foix are volcanic tuffs containing fragments of basalts, limestones and marls. The mean anisotropy for 11 rock cylinders is:

$$\text{maximum} : \text{intermediate} : \text{minimum} = 1{\cdot}024 : 1{\cdot}009 : 1$$

and the minimum susceptibility is $1\,550 \times 10^{-6}$ e.m.u./g. The directions of maximum susceptibility show no significant preferred orientation either before and after corrections have been applied for geological strike and dip. This is to be expected as the rock specimens are almost isotropic and the directions of their principal susceptibilities are therefore poorly determined.

At the second locality (south-east of Bayonne) samples were collected from a basaltic lava flow at two sites. Of the 18 samples, 13 were found to be reversely magnetized. The mean anisotropy of susceptibility for 15 rock cylinders is:

$$\text{maximum} : \text{intermediate} : \text{minimum} = 1{\cdot}065 : 1{\cdot}034 : 1$$

and the minimum susceptibility $\chi = 520 \times 10^{-6}$ e.m.u./g. Again, the directions of principal susceptibilities show no significant preferred orientation agreeing with the very small degree of anisotropy.

It is concluded that the pressures and temperatures associated with the Alpine folding did not cause a significant shape or crystallographic alignment of the ferromagnetic minerals.

5. Deccan Trap Rocks of India

Clegg and his co-workers in London have made a magnetic study of the Deccan Trap rocks of India. The rocks are mainly basaltic lavas and range in age from Eocene to possibly late Jurassic.

Studies of the natural remanent magnetization of these rocks have been described in Clegg & others (1956), Deutsch & others (1958) and Clegg & others (1958). Their results are summarized in stratigraphical order in Table 3.

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Table 3

Mean directions of natural remanent magnetization for Deccan Trap Rocks

Site	Locality	θ	ϕ
I	Upper Pavagadh rhyolite tuffs	355	+10
II–IV	Upper Nipani, Upper Amba and Lower Pavagadh	341	−24
V–VIII	Lower Nipani, Lower Amba, also Linga and Khandala	156	+56
IX	Rajmahal Traps (? Jurassic)	328	−64

θ = Declination measured clockwise from north.
ϕ = Inclination, with reference to the horizontal plane (positive inclination—downwards, negative inclination—upwards).

It is seen that the directions of natural remanent magnetization show a systematic decrease of inclination (ϕ) with altitude. Clegg & others (1958) give three possible causes for this. First, it may have occurred as a result of some physical phenomenon associated with different depths of burial such as magneto-striction (Graham 1957). Secondly, it may be due to partial magnetic instability which would tend to pull the directions towards the present field. This would tend to increase the inclinations of the reversely magnetized rocks (V–VIII) and decrease those of the normally polarized ones (I–IV and IX). Thirdly it may represent a continuous northward movement of India relative to the Equator during the time of eruption of the Deccan Traps.

For various reasons, Clegg & others conclude that the combined results for the Rajmahal and Deccan Traps conform best to the third postulate of a continuous northward movement of the Indian landmass from Jurassic to Cretaceous times.

To investigate further the first possibility, the susceptibility anisotropy has been measured for 26 specimens which represent a sampling throughout the whole succession of lavas. It was hoped to reveal any systematic variation of susceptibility anisotropy corresponding to the variation in magnetic inclination and if possible to relate this to the depth of burial.

The results are given in Table 4. The value of χ ranges from 285 to $3\,530 \times 10^{-6}$ e.m.u./g. The largest difference between maximum and minimum susceptibilities is found to be 3·6 per cent and the mean percentage difference between maximum and minimum susceptibilities is 1·6 per cent. Inspection of the ratios of the principal susceptibilities shows that the susceptibility is almost isotropic and the directions of the principal axes are therefore almost random.

It is concluded that the stresses associated with the depth of burial of the lavas were insufficient to cause shape or crystalline magnetic anisotropy and it is unlikely that the directions of natural remanent magnetization have been significantly affected by this cause.

6. Conclusions

Measurements for susceptibility anisotropy have been made for four sets of rocks. Two of these showed anisotropy whilst the other two were isotropic within the limits of the measuring apparatus. Of the two sets of anisotropic rocks, it is established that the anisotropy is due to different causes; to crystalline effect and shape effect respectively.

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Table 4

Susceptibility anisotropy for a selection of Deccan Trap specimens

Specimen Number	Anisotropy			Direction of Maximum k	
	Max. :	Int. :	Min.	θ	ϕ
A24	102·1	101·8	100	217	−35
K14a	102·1	100·2	100	134	−66
K29a	102·8	101·6	100	70	−78
K36aI	103·6	103·4	100	290	−41
K52a	103·5	102·2	100	347	−44
K56a	101·2	100·7	100	284	+37
K64aI	101·5	101·0	100	28	+28
K80aI	102·0	100·9	100	184	−55
K81a	101·4	101·0	100	131	−3
K82a	100·9	100·8	100	357	−37
K87a	100·8	100·4	100	159	−33
K95aI	100·7	100·3	100	152	−53
K92a	101·2	100·9	100	299	−50
K98aI	101·1	100·6	100	216	+8
K107aI	100·6	100·4	100	53	−16
K114a	101·0	100·8	100	32	−11
K116a	100·9	100·4	100	357	−66
K119a	102·3	101·1	100	156	+42
L49BI	101·2	100·3	100	32	+8
L58a	100·6	100·3	100	30	−29
L59a	101·9	101·6	100	42	+34
L64a	101·0	100·4	100	49	+34
L67a	101·0	100·5	100	70	−81
N34I	101·8	101·0	100	232	−31
R231	102·3	100·4	100	71	+4
R27	102·4	100·8	100	250	−13

θ = Declination of maximum susceptibility, measured clockwise from north.
ϕ = Inclination of maximum susceptibility with reference to the horizontal plane.

For the Welsh slates the maximum susceptibility, intermediate susceptibility and natural remanent magnetization all lie close to the cleavage plane. The susceptibility anisotropy is due to the preferred alignment of pyrrhotite grains which possess an easy plane of magnetization.

For the Skaergaard rocks, the maximum and intermediate susceptibilities lie close to the horizontal plane and the anisotropy is found to be due to the shape and distribution of isotropic titano-magnetite grains. The natural remanent magnetization of these rocks deviates from the ambient field but it is possible to determine accurately the direction of the past field from a knowledge of the susceptibility anisotropy and the directions of natural remanent magnetization. It is noted that anisotropy may be a factor contributing to the scatter in directions of natural remanent magnetization for some igneous rocks (Figure 3).

For the Pyrenees and Indian rocks, it is concluded that conditions of pressure and temperature during their history have been insufficient to cause susceptibility anisotropy and the natural remanent magnetization has been unaffected by this cause. This does not exclude the possibility that the remanent magnetization may have been affected by some other mechanism.

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In general, the fabric of igneous rocks will have been attained at temperatures higher than the Curie temperature of the ferromagnetic minerals which may be present. If susceptibility anisotropy exists, the natural remanent magnetization will contain a component of the ambient field and it is possible to determine the direction of the Earth's past field from a knowledge of the susceptibility anisotropy and natural remanent magnetization. Conversely, if a rock has been subject to temperatures higher than the Curie temperature of the minerals present during deformation, the natural remanent magnetization will contain a component of the Earth's field during the time of deformation. It is likely that similar considerations apply to chemical remanent magnetization.

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